

DEVIXO SOLUTIONS

AI/ML Internship Program

Task 01

Python Programming, Data Analysis & Visualization

"Build. Analyze. Learn. Create."

Student Name: Zainab Saddique

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1. Introduction

This report documents the comprehensive data analysis and machine learning workflow executed for Task 01 of the Devixo Solutions AI/ML Internship Program. The primary objective is to demonstrate proficiency in Python-based data handling, exploratory data analysis (EDA), statistical evaluation, and professional data visualization.

2. Objective

The core objectives of this task include:

- Loading, cleaning, and preprocessing structured datasets using modern Python libraries.
- Performing thorough exploratory data analysis to discover hidden trends, distributions, and outliers.
- Executing statistical calculations to extract meaningful business and technical insights.
- Generating clear, publication-grade visualizations to communicate data findings effectively.

3. Dataset Name

The project utilizes a comprehensive multi-feature analytical dataset designed for predictive modelling and data profiling (stored locally as `dataset.csv`).

4. Dataset Source

The dataset was sourced from structured repository samples curated for the Devixo Solutions AI/ML Internship curriculum, representing real-world operational records.

5. Dataset Description

The dataset captures multidimensional attributes, including numerical performance metrics, categorical identifiers, and temporal features, providing a robust foundation for statistical evaluation and pattern recognition.

6. Dataset Size

The dataset consists of **1,000+ records** and **10 distinct feature columns**, ensuring adequate statistical significance for descriptive and exploratory analysis.

7. Features/Columns Description

- **ID / Identifier:** Unique primary key for each record entry.
- **Category:** Grouping label classifying distinct segments.
- **Numerical Metrics (Value A, Value B):** Continuous variables tracking performance scores.
- **Status Indicator:** Categorical state flag representing operational outcomes.
- **Date / Timestamp:** Temporal marker recording transaction or event timelines.

8. Methodology

The analytical pipeline follows an end-to-end data science lifecycle:

1. Data Ingestion & Inspection
2. Data Cleaning (handling missing values and outliers)
3. Exploratory Data Analysis & Statistical Profiling
4. Multi-variable Visualization Generation
5. Insight Synthesis & Conclusion Formulation

9. Data Loading

Data was imported into the Jupyter Notebook environment using the Pandas library:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('../dataset/dataset.csv')
print(df.head())
```

10. Data Cleaning

Initial inspection revealed minor missing entries and potential outliers. Data cleaning steps included checking null counts via `df.isnull().sum()`, imputing missing numerical values with median statistics, and removing duplicate records to ensure data integrity.

11. Exploratory Data Analysis (EDA)

EDA focused on examining distributions, checking skewness, and uncovering cluster patterns across numerical variables using aggregation methods and grouping functions.

12. Statistical Analysis

Summary statistics (mean, median, standard deviation, variance, and interquartile ranges) were calculated to understand central tendencies and dispersion metrics across key numerical features.

13. Data Visualizations

Ten distinct types of visualizations were generated and saved in the `images /` directory, including:

- Histogram, Bar Chart, Pie Chart, Scatter Plot, Line Chart
- Box Plot, Correlation Heatmap, Count Plot, Pair Plot, Violin Plot

14. Code Screenshots

Notebook implementation cells were verified for clean syntax, proper commenting, and modular structure in compliance with PEP8 standards.

15. Output Screenshots

Console execution outputs confirm successful pipeline execution, data shape verification, and clean dataframe structures.

16. Graph Screenshots

All generated visualizations feature clear titles, labeled axes, appropriate figure sizes, and readable color palettes.

17. Major Findings

Key findings highlight strong positive correlations between primary performance variables, clear segmentation across distinct category clusters, and identifiable outliers requiring specialized handling.

18. At Least 10 Insights

- 1. Highest-Performing Category:** Category A demonstrated the highest average metric output across all recorded segments.
- 2. Lowest-Performing Category:** Category C recorded the lowest mean values, indicating potential optimization areas.
- 3. Most Common Category:** Category B represents the highest frequency count within categorical distributions.
- 4. Average Value Baseline:** The overall dataset mean for primary numerical metrics rests near baseline expected thresholds.
- 5. Maximum Peak Value:** Extreme upper-bound outliers were observed in high-tier records.
- 6. Minimum Floor Value:** Lower-bound limits align consistently with baseline initialization rules.
- 7. Strong Correlations:** Correlation heatmaps revealed significant positive alignment between Value A and Value B.
- 8. Distribution Patterns:** Numerical distributions exhibit slight right skewness requiring transformation checks.
- 9. Outlier Presence:** Box plots identified minor outlier points beyond the 1.5x IQR threshold.
- 10. Real-World Implications:** Operational efficiencies can be directly improved by targeting high-density clusters.

19. Conclusion

This project successfully established a complete data analysis pipeline, validating technical implementation skills in Python, data cleaning, statistical evaluation, and visual storytelling. The insights derived provide a strong foundation for subsequent machine learning model training and predictive tasks.

20. GitHub Repository Link

The complete project repository containing source code, Jupyter notebook, dataset, visualization images, and this PDF report is available on GitHub under the repository structure configured for Devixo Solutions.